Aeromonas hydrophila Intestinal Disease — Comprehensive Disease Characterization

MONDO ID: MONDO:0024459 | **Category:** Infectious Disease (bacterial gastroenteritis) | **Report date:** 2026-09-30

Summary

Aeromonas hydrophila intestinal disease is an acute, non-genetic infectious gastroenteritis caused by the Gram-negative, motile, waterborne/foodborne bacterium *Aeromonas hydrophila* (NCBI Taxonomy txid644). Clinically it spans a wide spectrum from mild self-limited watery diarrhea, through severe cholera-like "rice-water" diarrhea with dehydration, to dysentery-like bloody diarrhea. In a clinical series of 50 patients with *Aeromonas* acute gastroenteritis, 26% presented with cholera-like rice-water stools and severe dehydration, 16% with a dysentery-like picture, and one died of dehydration and septicaemia (PMID: 29205196). Because *Aeromonas* is recovered at similar rates from symptomatic and asymptomatic stools, its status as a true enteric pathogen versus a commensal/coloniser remains debated (PMID: 39388901).

The disease is **toxin-driven and multifactorial**. Ingestion is followed by chemotaxis and adhesin/pilus-mediated attachment to intestinal mucus, then secretion and proteolytic activation of pore-forming toxins — principally aerolysin (*aer/aerA*) and the cytotoxic enterotoxin **Act** (*act*) — supplemented by hemolysins, cytotonic enterotoxins (*alt/ast*), and proteases. Act forms transmembrane β-barrel pores, activates p38/JNK/ERK MAPK cascades, drives COX-2/PGE2 and pro-inflammatory cytokine production (TNF-α, IL-1β, IL-6), raises cAMP, and induces caspase-dependent apoptosis — together explaining both the secretory (cholera-like) and inflammatory/dysenteric phenotypes. Virulence scales with the number of toxin genes a strain carries.

The condition is usually **self-limited** and managed with rehydration; when antimicrobials are needed, intrinsic β-lactamase–mediated resistance to ampicillin/penicillins means treatment favours fluoroquinolones, 3rd/4th-generation cephalosporins, carbapenems, aminoglycosides, and cotrimoxazole. *A. hydrophila* is also a **major One Health pathogen**, causing Motile Aeromonad Septicaemia (MAS) in farmed fish with heavy economic losses; here prevention is achieved at the reservoir level through vaccines, phytobiotics, and probiotics (including engineered antimicrobial-peptide–secreting strains) rather than any licensed human vaccine. Global pooled prevalence in children with diarrhea is **4.2% (95% CI 3.1–5.6%)**, higher where water and sanitation are poor, and declining over time (PMID: 37217902).

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1. Disease Information

Overview. *Aeromonas hydrophila* intestinal disease is an acute bacterial gastroenteritis caused by an environmental, oxidase-positive, facultatively anaerobic, motile Gram-negative rod belonging to the family *Aeromonadaceae*. It is classed as an emerging, opportunistic, waterborne/foodborne enteric infection. The clinical presentation ranges from mild watery diarrhea to cholera-like secretory diarrhea and dysentery-like bloody diarrhea (PMID: 29205196).

Key identifiers. - **MONDO:** MONDO:0024459 - **MeSH:** *Aeromonas* infections / Gram-Negative Bacterial Infections; organism MeSH "Aeromonas hydrophila" - **NCBI Taxonomy:** *Aeromonas hydrophila* = txid644; genus *Aeromonas* = txid642 - **ICD-10:** A04.8 (Other specified bacterial intestinal infections) — *Aeromonas* enteritis has no dedicated code and is captured under other/specified bacterial intestinal infections. - **OMIM / Orphanet:** Not applicable — this is an acquired infectious disease with no Mendelian basis; no OMIM or Orphanet rare-disease entry.

Synonyms / alternative names. *Aeromonas* gastroenteritis; *Aeromonas*-associated diarrhea; *Aeromonas* enteritis; mesophilic *Aeromonas* infection; motile aeromonad infection (in veterinary contexts, Motile Aeromonad Septicaemia, MAS, when systemic in fish).

Information source. Content is aggregated from disease-level resources — clinical case series, case-control studies, PCR/genomic surveys, and systematic reviews/meta-analyses — rather than individual EHR-derived patient records.

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2. Etiology

Primary cause — infectious. The disease is caused by ingestion of *A. hydrophila* (and closely related motile mesophilic aeromonads such as *A. caviae*, *A. veronii*, *A. dhakensis*). There is **no host genetic etiology**; the disease is entirely acquired. Virulence is multifactorial and strain-dependent (Findings F002, F003).

Environmental / risk factors. - **Waterborne and foodborne transmission** — *Aeromonas* is transmitted via contaminated water, fish, and chicken meat, and can grow competitively at 5 °C, making it a refrigerated-food hazard (PMID: 27051177). - **Young age** — isolation rates are significantly higher in diarrheal than control children ($p<0.001$) (PMID: 15005966); *A. hydrophila* was among the most frequent enteropathogens (9%) in Somali children (PMID: 3076719). - **Immunocompromise / malignancy** — in immunocompromised children with diarrhea (mostly acute lymphoblastic leukemia on chemotherapy), *Aeromonas* spp. were the most commonly detected target (14.6%), with bloody diarrhea in 98.5% (PMID: 38778271). - **Advanced age / hematologic disease** — severe/hemorrhagic disease reported in elderly patients with aplastic anemia/thrombocytopenia (PMID: 41903649). - **Poor water and sanitation** — prevalence 8.8% where water/sanitation quality score is <25% (PMID: 37217902).

Protective factors. No genetic protective variants are defined (acquired disease). Environmental/behavioural protection derives from **clean water, sanitation, and food safety** (adequate cooking, avoiding prolonged refrigerated storage). In aquaculture reservoirs, probiotics and phytobiotics are protective (Findings F005, F012).

Gene-environment interactions. Not applicable to the human host. The relevant axis is on the **pathogen side**: strains carrying more virulence genes are more lethal, and the environmental/host niche selects for hypervirulent clonal lineages (Findings F002, F007).

3. Phenotypes

Phenotype	Type	Characteristics	Frequency	Suggested HPO
Watery diarrhea (secretory, cholera-like)	Symptom	Acute onset; can cause severe dehydration	26% had rice-water stools (PMID: 29205196)	HP:0002014 (Diarrhea)
Dysentery / bloody diarrhea	Sign	Inflammatory, mucoid/ bloody stool	16% dysentery-like (PMID: 29205196); 98.5% bloody in immunocompromised children (PMID: 38778271)	HP:0025085 (Bloody diarrhea)
Dehydration	Sign	Can be severe; life-threatening	Common in severe cases	HP:0001944 (Dehydration)
Abdominal pain / cramps	Symptom	Accompanies diarrhea	Frequent	HP:0002027 (Abdominal pain)
Fever	Sign	Variable	Variable	HP:0001945 (Fever)
Vomiting / nausea	Symptom	Variable	Variable	HP:0002013 (Vomiting)
Sepsis / bacteremia	Outcome	Rare complication, mainly immunocompromised/ elderly	Rare; one death in 50-patient series	HP:0100806 (Sepsis)

Onset: acute (hours to days after exposure). **Severity:** variable — mild and self-limited in most immunocompetent hosts, severe/hemorrhagic in the immunocompromised and elderly. **Progression:** typically self-limited over days; episodic depending on re-exposure. **Quality-of-life impact:** for uncomplicated cases, transient; severe dehydration or sepsis can be life-threatening.

4. Genetic/Molecular Information

Not applicable to the human host. This is an infectious disease; there are **no causal host genes, pathogenic germline/somatic variants, modifier genes, host epigenetic changes, or chromosomal abnormalities**. Host genetic testing is not indicated.

The relevant "genetics" is **microbial virulence genetics**: - **aerolysin** (*aer/aerA*) — β-pore-forming toxin, present in up to 87% of isolates (PMID: 25216211); 72.7% of fish isolates (PMID: 30989358). - **cytotoxic enterotoxin** (*act*) — 7.7–18.2% of isolates. - **cytotonic enterotoxins** (*alt*, *ast*) — *ast* 17% (PMID: 25216211). - **hemolysin A** (*hylA*), **serine protease** (*ahp*), **temperature-sensitive protease** (*eprCAI*). - **Regulation:** DNA adenine methyltransferase (Dam) and tRNA-modifying enzyme GidA co-regulate *act* expression via GATC methylation (PMID: 22391092). - **Pfam / protein family:** aerolysin/ETX family, **Pfam PF01117** (aerolysin).

Strains carrying more virulence genes show progressively higher proteolytic, hemolytic, and cytotoxic activity and lower LD50 in zebrafish; the *aer*+*alt*+*act*+*eprCAI*+*ahp*+ genotype is enriched among diseased hosts and most virulent (PMID: 21219556, PMID: 22725694).

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5. Environmental Information

- **Environmental reservoir:** freshwater, brackish water, wastewater, aquaculture ponds, and food (fish, poultry). Wastewater treatment plants are reservoirs for antibiotic-resistance gene dissemination in *Aeromonas* (PMID: 40269715).
- **Psychrotrophy:** growth at 5 °C makes refrigerated foods a hazard (PMID: 27051177).
- **Lifestyle factors:** consumption of untreated water, raw/undercooked fish and shellfish; exposure to recreational or occupational water (e.g., paddy-field injuries associated with invasive *Aeromonas* disease, PMID: 42288750).
- **Infectious agents:** *Aeromonas hydrophila* (txid644); related *A. caviae*, *A. veronii*, *A. dhakensis*, *A. enteropelogenes*. Toxin entities: aerolysin, cytotoxic enterotoxin Act.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating exposure → clinical manifestation)

1. **Ingestion** of *A. hydrophila* in contaminated water/food **leads to** delivery of viable, motile bacteria to the small and large intestine.
2. **Chemotaxis toward intestinal mucus** (flagella-driven) **results in** localization at the mucosal surface; chemotaxis may be required for obligate pathogenicity (PMID: 18355183).
3. **Adhesin/pilus-mediated attachment** to mucus glycoproteins (galactose/mannose receptors) **leads to** stable colonization; stronger adhesion correlates with greater cytotoxicity (PMID: 18924378, PMID: 1363704).
4. **Secretion of toxins** (T2SS-secreted Act and aerolysin; hemolysins; proteases) **results in** inactive pro-toxins at the epithelial surface.
5. **Proteolytic activation** (C-terminal propeptide cleavage of proaerolysin/Act) **leads to** active toxin monomers with increased β-sheet content (PMID: 9398250).
6. **GPI-anchored receptor binding** + **heptamerization results in** a prepore that undergoes a concerted "swirling" collapse, **inserting a transmembrane β-barrel pore** (1.14–2.8 nm) (PMID: 23912165, PMID: 9317040, PMID: 34162996).
7. **Membrane permeabilization** branches into two downstream arms:
8. **7a (secretory arm):** pore formation + Act signalling **raises cAMP and activates COX-2/PGE2 and eicosanoid metabolism** → net fluid/electrolyte secretion → **watery, cholera-like diarrhea** (PMID: 10768977).
9. **7b (inflammatory/cytotoxic arm):** Act activates **p38, JNK, ERK1/2 MAPK cascades**, induces **TNF-α, IL-1β, IL-6, iNOS**, and triggers **caspase-3/8/9–dependent apoptosis** in macrophages and human intestinal epithelial cells (T84, HT-29) → mucosal injury, neutrophilic inflammation → **dysenteric/bloody diarrhea** (PMID: 15215244).
10. **(Inferred, strain-dependent)** In strains with T3SS/T6SS, effectors **AexU (T3SS) and Hcp (T6SS)** subvert host immune signalling and promote apoptosis/pyroptosis **leading to** deeper invasion and rarely **bacteremia/sepsis** (PMID: 24199174, PMID: 23611906).
11. **(Inferred, lineage-dependent)** Epidemic ST251 clones achieve high virulence **independent of** classical T3SS/T6SS/lateral flagella, instead exploiting **myo-inositol, sialic acid, and L-fucose utilization** pathways (PMID: 26014286).

Category checklist mapped to the chain

- **Molecular pathways:** MAPK (p38/JNK/ERK); cAMP/PKA; arachidonic-acid/COX-2–PGE2 eicosanoid pathway; caspase apoptotic cascade.
- **Cellular processes:** pore-mediated membrane permeabilization; apoptosis (GO:0006915); inflammation; cytokine production (GO:0001816).
- **Protein dysfunction (pathogen toxins):** proaerolysin → active heptameric β-barrel pore; random-coil→β-sheet transition.
- **Immune involvement:** pro-inflammatory cytokines (TNF-α/IL-1β/IL-6), iNOS induction, neutrophilic infiltration; immune subversion by secretion-system effectors.
- **Tissue damage mechanisms:** epithelial pore lysis, apoptosis/necrosis, inflammatory injury.
- **Biochemical abnormalities:** dysregulated ion/fluid transport (secretory diarrhea), elevated cAMP.

Suggested GO terms: GO:0006915 (apoptotic process), GO:0000165 (MAPK cascade), GO:0006954 (inflammatory response), GO:0019836 (hemolysis in another organism), GO:0140911 (pore-forming activity), GO:0001816 (cytokine production). **Suggested CL terms:** CL:0000584 (enterocyte), CL:0002253 (epithelial cell of large intestine), CL:0000235 (macrophage), CL:0000775 (neutrophil).

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7. Anatomical Structures Affected

- **Primary organ:** intestine — small intestine and colon (**UBERON:0000160 intestine; UBERON:0002108 small intestine; UBERON:0001155 colon**). Body system: digestive/gastrointestinal (**UBERON:0001007 digestive system**).
- **Secondary involvement:** bloodstream (bacteremia/sepsis) in immunocompromised/elderly hosts; rarely liver (acute hepatitis with invasive *Aeromonas*, PMID: 41356972).
- **Tissue/cell level:** intestinal **epithelium (mucosa)** — enterocytes (CL:0000584) and colonic epithelial cells (CL:0002253) are primary targets; **macrophages (CL:0000235)** and **neutrophils (CL:0000775)** mediate inflammation. Toxins act on GPI-anchored-protein–bearing membranes.
- **Subcellular level:** plasma membrane (pore insertion; **GO:0005886**); mitochondria/cytoplasm (caspase-9 intrinsic apoptosis; **GO:0005739**); cell-surface GPI anchor (**GO:0098552**).

- **Localization / lateralization:** diffuse mucosal involvement; not lateralized.

8. Temporal Development

- **Onset:** acute, within hours to a few days of ingestion (not congenital, not insidious).
- **Progression:** most cases self-limited over several days; a minority (severe cholera-like) progress rapidly to dehydration; dysenteric cases run an inflammatory course. No chronic-lifelong course in immunocompetent hosts.
- **Disease course pattern:** acute, self-limited; can be episodic with re-exposure. Rare progression to invasive/systemic disease in vulnerable hosts.
- **Remission:** usually spontaneous with supportive rehydration; antibiotics shorten severe cases.
- **Critical periods:** early rehydration is the key intervention window; in severe/immunocompromised cases, early appropriate antibiotics prevent progression to sepsis.

9. Inheritance and Population

- **Inheritance:** Not applicable — acquired infectious disease, no heritability, penetrance, expressivity, anticipation, founder effects, consanguinity, or carrier frequency.
- **Epidemiology:** Global pooled prevalence in children with diarrhea = **4.2% (95% CI 3.1–5.6%)** across 31 studies / 38,663 participants (PMID: 37217902). Higher in upper-middle-income countries (5.1%), very populous countries (>100 million: 9.4%), and poor-sanitation settings (<25% score: 8.8%); a significant **decreasing trend over time** (P=0.0001).
- **Demographics:** disproportionately affects **young children**, the **elderly**, and the **immunocompromised** (Finding F008). Higher isolation in diarrheal vs control children ($p<0.001$, PMID: 15005966).
- **Geographic distribution:** worldwide; higher burden where water/sanitation is poor; endemic in tropical/subtropical aquaculture regions. Epidemic hypervirulent clones (ST251) reported from China and the southeastern USA (PMID: 26014286).
- **Seasonality:** peaks in warm months.

10. Diagnostics

- **Stool culture** on blood/selective agar with **oxidase-positive** screening is the mainstay.
- **Species-level identification requires molecular methods** — biochemical and automated systems misidentify *Aeromonas*. In one case, MALDI-TOF gave high scores for two species and an automated system called it "*A. veronii* complex"; **definitive ID required multilocus phylogenetic analysis of *gyrB, rpoD, gyrA*** (PMID: 41903649).
- **PCR / molecular:** multiplex PCR for virulence genes (*aer/act/alt/ast*), and *gyrB*/16S rDNA sequencing for detection and genotyping (PMID: 25216211, PMID: 22725694).
- **Biosensors:** DNA hybridization electrochemical biosensors developed for *A. hydrophila* (PMID: 20829023).
- **Clinical criteria / differential diagnosis:** acute infectious diarrhea; distinguish from *Vibrio cholerae* (secretory), *Shigella*/*Campylobacter*/*Salmonella*/EIEC (dysenteric), *C. difficile* (antibiotic-associated). Stool culture is important in severe or protracted illness when antimicrobial therapy is contemplated (PMID: 3683260).
- **Genetic/omics diagnostics:** host genetic testing not applicable. Pathogen WGS/MLST is used for outbreak tracing and AMR profiling.
- **Screening:** no human screening program; environmental/food surveillance applies.

11. Outcome / Prognosis

- **Prognosis:** generally good; most cases are **self-limited** in immunocompetent hosts.
- **Mortality:** low overall but non-zero — one death (dehydration + septicaemia) in a 50-patient series (PMID: 29205196). Invasive/bacteremic disease carries substantial mortality, especially in immunocompromised/elderly patients (PMID: 41903649, PMID: 42288750).
- **Complications:** severe dehydration, dysentery, bacteremia/sepsis, rarely acute hepatitis; carbapenem treatment failure reported in critically ill patients due to PK/PD factors (PMID: 42288750).
- **Prognostic factors:** host immune status (immunocompromise, malignancy, advanced age worsen outcomes); strain virulence-gene load and hypervirulent lineage; timeliness of rehydration and appropriate antibiotics; AMR profile (ESBL/integron-borne resistance worsens outcomes, PMID: 41625157).

12. Treatment

- **Supportive care (cornerstone):** oral/IV **rehydration** and electrolyte correction; most immunocompetent cases need no antibiotics. **NCIT:** Fluid and Electrolyte Therapy.
- **Antimicrobials (when severe/invasive or immunocompromised):**
- **Avoid** ampicillin/amoxicillin and 1st-generation cephalosporins — *A. hydrophila* is **intrinsically resistant** via chromosomal β-lactamases (only ~14% amoxicillin-susceptible) (PMID: 29205196).
- **Effective agents:** fluoroquinolones (ciprofloxacin), 3rd/4th-generation cephalosporins (ceftazidime, cefepime), carbapenems (meropenem), aminoglycosides (amikacin, gentamicin), aztreonam, azithromycin, cotrimoxazole — gastroenteritis isolates are almost uniformly susceptible (PMID: 38783344). **NCIT:** Ciprofloxacin, Ceftazidime, Meropenem, Amikacin, Trimethoprim-Sulfamethoxazole.
- **AMR caveat:** emerging ESBL (CTX-M) and class-1-integron–borne resistance, including carbapenem treatment failures (PMID: 42288750, PMID: 41625157). Susceptibility testing should guide therapy.
- **Advanced/experimental therapeutics:** none specific for human *Aeromonas* enteritis. In aquaculture, engineered probiotics secreting antimicrobial peptides are investigational (see Prevention).

13. Prevention

- **Primary prevention:** clean water, sanitation, and food safety (adequate cooking; limiting refrigerated storage given growth at 5 °C) (PMID: 27051177, PMID: 37217902).
- **Human immunization: no licensed human vaccine.**
- **Veterinary/aquaculture prevention (One Health):**
- **Vaccines:** whole-cell inactivated *A. hydrophila* vaccines give the highest protection (vs live-attenuated/recombinant); i.p. injection most effective; feed-based bivalent (*S. iniae* + *A. hydrophila*) vaccines raise IgM and improve survival (PMID: 32331001, PMID: 33857622, PMID: 32615969).
- **Probiotics:** dietary *Saccharomyces cerevisiae* cut post-challenge mortality from 53.3% to 13.3% in African catfish (PMID: 41946979); *Bacillus licheniformis*, *Lactobacillus brevis* (+trans-cinnamic acid), and engineered *Bacillus subtilis* secreting piscidin-1/hepcidin

improved survival (PMID: 42067537, PMID: 42014760, PMID: 42176787); *Clostridium butyricum* blocks colonization in vitro (PMID: 18795975).

- **Phytobiotics:** dietary essential oils (basil, thyme, fenugreek) improve immunity and resistance (PMID: 42400869, PMID: 42376504).
- **Public health:** integrated One Health AMR surveillance across environmental, food, and clinical compartments (PMID: 42288750).

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14. Other Species / Natural Disease

- **Taxonomy of pathogen:** *Aeromonas hydrophila* NCBI **txid644**; genus *Aeromonas* txid642.
- **Natural disease — fish:** *A. hydrophila* is the usual etiological agent of **Motile Aeromonad Septicaemia (MAS)** in tilapia (*Oreochromis niloticus*), carp (*Cyprinus carpio*), catfish (*Clarias gariepinus*), and other freshwater fish, with high mortality and major economic loss (PMID: 32331001, PMID: 30989358).
- **Other hosts / zoonotic potential:** environmental organism of aquatic habitats; found in hippopotamus oral flora, of zoonotic importance in bite-wound infections (PMID: 40269996). Human infection is acquired from environmental/animal reservoirs (One Health).
- **Comparative biology:** the same toxin/virulence machinery (Act, aerolysin, T3SS/T6SS) drives disease across fish and mammals, and virulence-gene load predicts lethality across host models (zebrafish, mice) — indicating conserved mechanisms.

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15. Model Organisms

Model	Type	Use / recapitulation	Reference
Zebrafish (*Danio rerio*)	Vertebrate, in vivo	LD50 challenge; virulence-gene load correlates with lethality; probiotic protection	PMID: 21219556, PMID: 42176787
Mouse	Mammalian, in vivo	Lethality of wound pathotype E1 vs environmental ATCC 7966; IV LD50 of Act = 27.5 ng	PMID: 23611906, PMID: 9317040
Rat ileal loop	Mammalian, ex vivo	Minimal fluid-secretion dose of Act = 200 ng — models secretory diarrhea	PMID: 9317040
Tilapia / carp / catfish	Vertebrate, in vivo	MAS disease model; vaccine and probiotic efficacy testing	PMID: 32331001, PMID: 41946979
Human intestinal epithelial lines (T84, HT-29)	In vitro	Act-induced MAPK activation and apoptosis	PMID: 15215244
Murine macrophages	In vitro	Act-induced cytokine/eicosanoid production, apoptosis	PMID: 10768977
Rabbit intestine	Ex vivo	Pili adhesion / colonization-factor studies	PMID: 1363704
HEK293 GPI-biosynthesis knockout library	In vitro	Defines aerolysin GPI-anchor receptor requirements	PMID: 34162996

Model limitations: no single model captures the full human clinical spectrum plus the pathogen-vs-commensal ambiguity; fish MAS is a systemic septicaemia rather than a purely enteric disease.

Key Findings (with statistical evidence)

F001 — Clinical spectrum and pathogen-vs-commensal ambiguity

A. hydrophila causes acute gastroenteritis ranging from watery/cholera-like to dysenteric diarrhea, but is often a minor contributor or coloniser. In 50 patients, 26% had cholera-like rice-water stools with severe dehydration, 16% a dysentery-like picture, and one died (PMID: 29205196). *Aeromonas* was among the most frequent enteropathogens (9%) in Somali children (PMID: 3076719). In N. Israel, positivity was similar in symptomatic (4.24–4.81%) and asymptomatic (4.9%) stools — *"similar positivity rates of Aeromonas spp. in symptomatic and asymptomatic making the distinction between true pathogen versus commensal bacteria difficult"* (PMID: 39388901).

F002 — Multifactorial, dose-dependent virulence

Virulence combines aerolysin (*aer*), Act (*act*), hemolysins, cytotonic enterotoxins (*alt/ast*), and proteases. *aer* was present in 87%, *act* 7.7%, *ast* 17% of a Libyan isolate set (PMID: 25216211). Crucially, *"as the present of virulence genes increased, the proteolytic, haemolytic and cytotoxic activities of extracellular products also increased"* (PMID: 21219556), and 5-gene strains had lower LD50 in fish (PMID: 22725694).

F003 — Act is the core enterotoxic engine

Act is a 52-kDa aerolysin-related pore-forming toxin (pores 1.14–2.8 nm; rat ileal-loop minimal dose 200 ng; mouse IV LD50 27.5 ng) (PMID: 9317040). It *"activates stress-associated p38, c-Jun NH(2)-terminal kinase (JNK), and extracellular signal-regulated kinase 1/2 (ERK1/2)"* and induces apoptosis (PMID: 15215244), and *"stimulated the production of tumor necrosis factor alpha and upregulated ... IL-1beta ... and IL-6"* (PMID: 10768977).

F004 — Intrinsic β-lactam resistance shapes therapy

Isolates are uniformly susceptible to cefepime, amikacin, azithromycin, meropenem; only 14% to amoxicillin (PMID: 29205196); gastroenteritis isolates almost uniformly susceptible to amikacin, gentamicin, aztreonam, cefepime, ceftazidime, ciprofloxacin, meropenem (PMID: 38783344).

F005 — Major veterinary disease (MAS); vaccines protect

"The etiological agent of this disease is usually Aeromonas hydrophila. Mortality rates due to MAS are frequently high" (PMID: 32331001); whole-cell inactivated vaccines give highest protection.

F006 — Molecular ID required for accurate speciation

Routine MALDI-TOF/automated systems misidentify *Aeromonas*; definitive ID needs *gyrB/rpoD/gyrA* MLPA (PMID: 41903649).

F007 — Hypervirulent lineages independent of classical secretion systems

For epidemic ST251 clones, *"the known virulence factors, including a type III secretion system, a type VI secretion system and lateral flagella, are not required for the high virulence"* — instead myo-inositol/sialic-acid/L-fucose metabolism is implicated (PMID: 26014286).

F008 — Waterborne/foodborne transmission; vulnerable hosts

"The ability of these bacteria to grow competitively at 5°C may be indicative of their potential as a public health hazard" (PMID: 27051177); enriched in diarrheal children ($p<0.001$, PMID: 15005966); leading enteric agent (14.6%) in immunocompromised children (PMID: 38778271).

F009 — Aerolysin: prototype β-pore-forming toxin

Secreted as proaerolysin, proteolytically activated (*"a decrease in random structure and an increase in beta-sheet content"* PMID: 9398250), binds GPI anchors, heptamerizes (mass 333,850 Da confirms heptamer, PMID: 8617369), and inserts via a *"unique concerted swirling movement ... leading to the insertion of a transmembrane β-barrel"* (PMID: 23912165).

F010 — Global prevalence 4.2%, sanitation-linked, declining

"The pooled prevalence of Aeromonas in children with diarrhoea worldwide was 4.2% (95% CI 3.1-5.6%)"; poor sanitation → 8.8%; decreasing trend over time (PMID: 37217902).

F011 — Adhesion/colonization gates cytotoxicity

"bacteria that showed a stronger adhesion to intestinal mucus were cytotoxic ... and bacteria that showed a weaker adhesion ... did not lead to alterations of monolayers" (PMID: 18924378); pili are colonization factors via galactose/mannose receptors (PMID: 1363704).

F012 — Antibiotic-alternative prevention works in aquaculture

Dietary probiotics reduced post-challenge mortality from 53.3% to 13.3% (PMID: 41946979); rising AMR *"necessitat[es] the urgent need for sustainable alternative treatments"* (PMID: 42573755).

Mechanistic Model (synthesis)

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Ingestion (contaminated water/food, grows at 5°C)
       │
       ▼
Chemotaxis (flagella) ──► Adhesion (pili, mucus glycoprotein receptors)
       │                        │  stronger adhesion → greater cytotoxicity
       ▼                        ▼
Toxin secretion (T2SS: Act, aerolysin; hemolysins; proteases)
       │
       ▼
Proteolytic activation (propeptide cleavage; coil→β-sheet)
       │
       ▼
GPI-receptor binding → heptamerization → "swirling" β-barrel pore insertion
       │
       ├──────────────────► SECRETORY ARM: ↑cAMP, COX-2/PGE2, eicosanoids
       │                      → fluid/electrolyte secretion → CHOLERA-LIKE DIARRHEA
       │
       └──────────────────► INFLAMMATORY ARM: p38/JNK/ERK MAPK, TNF-α/IL-1β/IL-6,
                             iNOS, caspase-3/8/9 apoptosis
                             → mucosal injury + neutrophilic inflammation
                             → DYSENTERIC/BLOODY DIARRHEA
       │
       └── (strain-dependent) T3SS/AexU + T6SS/Hcp immune subversion → INVASION/SEPSIS
       └── (lineage-dependent, ST251) metabolism-linked hypervirulence (inositol/sialic acid/fu
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Upstream events (colonization, adhesion, toxin activation) gate **downstream** effects (pore formation, signalling, cell death). The two clinical phenotypes (secretory vs dysenteric) branch from the same pore-forming event depending on the balance of cAMP/eicosanoid secretion versus MAPK-driven apoptosis/inflammation.

Evidence Base (selected)

PMID	Title (abbrev.)	Supports
29205196	Speciation, clinical profile & antibiotic resistance	Clinical spectrum (F001); intrinsic β-lactam resistance (F004)
21219556	*alt/ahp/aerA* virulence & zebrafish virulence	Dose-dependent virulence (F002)
9317040	Act hyperproduction & mechanism	Act pore-forming mechanism (F003)
15215244	Act activates MAPK & apoptosis	Inflammatory arm (F003)
10768977	Act cytokines & arachidonic acid	Secretory/inflammatory cytokines (F003)
23912165	Aerolysin swirling pore assembly	Structural mechanism (F009)
8617369	Heptameric pore MALDI-TOF	Heptamer stoichiometry (F009)
37217902	Global prevalence meta-analysis	Epidemiology (F010)
18924378	Mucus adhesion & cytotoxicity	Colonization step (F011)
26014286	ST251 comparative genomics	Lineage-specific virulence (F007)
32331001	Tilapia MAS vaccination review	Veterinary disease & prevention (F005)
41946979	Dietary yeast vs challenge	Probiotic prevention (F012)
41903649	Diagnostic pitfalls case report	Molecular ID requirement (F006)
27051177	Foodborne review	Transmission & psychrotrophy (F008)

Challenging / nuancing evidence: PMID: 3584426 found no correlation between toxin production and gastroenteritis, and PMID: 39388901 shows equal symptomatic/asymptomatic carriage — both underscore that *Aeromonas* enteropathogenicity is strain-dependent and sometimes commensal, tempering simple causal claims.

Limitations and Knowledge Gaps

1. **Causality unmet at population level.** Similar carriage in symptomatic and asymptomatic individuals (PMID: 39388901) and no toxin–disease correlation in one series (PMID: 3584426) mean attribution specifically to *A. hydrophila* is often uncertain; co-infections are common.
2. **Species misidentification** historically conflated *A. caviae*, *A. veronii*, *A. dhakensis*, and *A. hydrophila*, so species-specific clinical data are imprecise; *A. caviae* now dominates many clinical series (PMID: 38783344).
3. **Human vs fish evidence imbalance.** Much mechanistic and prevention data come from fish/aquaculture and cell/animal models, not human enteric disease.
4. **No human clinical trials** define antibiotic benefit for uncomplicated cases; treatment recommendations are extrapolated from susceptibility data.
5. **AMR evolution** (ESBL/CTX-M, class-1 integrons, carbapenem failures) is dynamic and may outpace current guidance.
6. **No host-genetic dimension** — sections 4 (host genetics) and 9 (inheritance) are inherently not applicable for an acquired infection.

Proposed Follow-up Actions

1. **Prospective, culture-independent, species-resolved cohort studies** (metagenomics + strain WGS) pairing symptom status with virulence-gene content to resolve the pathogen-vs-commensal question and define an attributable fraction.
2. **Randomized controlled trials** of antibiotic therapy vs rehydration alone in moderate–severe *Aeromonas* gastroenteritis, stratified by host immune status.
3. **Integrated One Health AMR surveillance** linking environmental/wastewater, food, aquaculture, and clinical isolates to track ESBL/integron gene flow (PMID: 42288750, PMID: 40269715).
4. **Mechanistic dissection of ST251-type metabolism-linked hypervirulence** (inositol/sialic acid/fucose utilization) as potential anti-virulence targets (PMID: 26014286).
5. **Anti-virulence / anti-toxin therapeutics** (Act/aerolysin pore inhibitors, adhesion blockers) and continued development of aquaculture probiotic/vaccine reservoir control to reduce human exposure.

6. **Standardize molecular species identification** (*gyrB/rpoD/gyrA*) in diagnostic labs to improve epidemiological precision.

Report compiled from 12 confirmed findings and 64 reviewed papers over 5 investigation iterations. Evidence types: human clinical series/meta-analysis, model organism (zebrafish, mouse, fish), in vitro (epithelial/macrophage cell lines), and structural/computational studies.
